BANNARI AMMAN INSTITUTE OF TECHNOLOGY INSTITUTE OF TECHNOLOGY Stay Ahead	**BANNARI AMMAN INSTITUTE OF TECHNOLOGY** (An Autonomous Institution Affiliated to Anna University, Chennai) SATHYAMANGALAM - 638 401
	Approved Questions S2 Semester
22MA201 – ENGINEERING MATHEMATICS II	

Q. No.	Questions
	Unit 1
1.	In an electric vehicle battery cooling system, the temperature T satisfies $\frac{dT}{dt} = -k(T - T_a)$, where T_a is ambient temperature. **Solve using separation of variables and evaluate** the time required for the battery to reach a safe temperature given initial conditions.
2.	In a fluid drainage system, the height h of water in a tank satisfies $\frac{dh}{dt} = -k\sqrt{h}$ **Apply separation of variables to compute** the time taken for the tank to empty and **interpret** how system parameters affect drainage rate.
3.	In a wireless sensor system, signal strength S decays as $\frac{dh}{dt} = -k\sqrt{h}$ **Solve the equation using separation of variables and analyze** how initial signal strength influences transmission range.
4.	In a population model of bacteria growth in a bioreactor,dP/dt=kP **Solve using separation of variables and identify** how doubling time depends on growth rate.
5.	A student solves dx/dy=xy but separates variables incorrectly. **Identify the correct separation step and compute** the general solution.
6.	In a heat transfer model, dT/dt=−kT. **Identify whether the equation is separable and justify.**
7.	For the equation dy/dx=x/y, **separate the variables and state** the integrable form.
8.	In an electric vehicle, the battery pack experiences non-uniform heating during fast charging. The temperature rise is influenced by internal resistance and external cooling, leading to the model $\frac{dy}{dx} + 3y = 2x - 1$ where y represents temperature deviation and x represents time. **Find** the temperature distribution using Leibnitz method and **analyze** whether the system reaches thermal stability.

9.	In a robotic arm, position error is governed by feedback and actuator delay. The system is modeled as $x\frac{dy}{dx} + y(x+1) = 1$ where y is error and x is time. **Convert**, **solve**, and **evaluate** whether the system eliminates steady-state error.
10.	A damping system in machinery follows $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$ An engineer incorrectly assumes integrating factor e^{-2x}, leading to wrong results. **Identify the mistake**, **compute correct integrating factor**, and **solve** the equation.
11.	In a smart sensor, voltage output varies with environmental conditions and is modeled as $(x+1)\frac{dy}{dx} - y = e^{3x}(x+1)^2$ **Transform** into standard form and **compute** integrating factor.
12.	A communication system processes signals following $\frac{dy}{dx} + \frac{y}{x\log x} = \frac{2\log x}{x\log x}$ **Evaluate** integrating factor and **interpret** the substitution used.
13.	In an RC circuit, voltage variation is modeled by $\frac{dy}{dx} + \frac{y}{x} = x^2$ **Identify** integrating factor and **state** its significance.
14.	In a chemical process, concentration change follows $\frac{dy}{dx} + 5y = 4x - 1$ **Find** integrating factor and **interpret** its role in simplifying solution.
15.	In an autonomous vehicle navigation system, the trajectory is represented by a family of curves $x^2 + y^2 = r^2$ where r varies depending on turning radius. **Form** the differential equation representing this motion and **interpret** its physical meaning.
16.	A vibrating system is described by displacement $y = a\sin(x+b)$ where a and b depend on system calibration. **Form** the differential equation by eliminating constants and **evaluate** its engineering significance.

17.	A signal amplification system follows $y = ae^{4x} + be^{-x}$where a, b depend on initial signal conditions. **Form** the differential equation and **analyze** system response behavior.
18.	A robot moves in straight lines described by $y = mx + c$ where slope varies with terrain. **Form** differential equation when both m and c are arbitrary and **justify** order.
19.	Fluid flow velocity is given by $y = \frac{a}{x} + b$ where constants depend on system pressure. **Form** the differential equation and **interpret** the result.
20.	A sensor output follows $y = e^{x}\,(acosx + bsinx\,)$. **Identify** the order of resulting differential equation and **justify**.
21.	A sensor output follows $y = e^{\mathrm{bx}}\left(acosx\right)$ **Identify** the order of resulting differential equation and **justify**.
22.	An EV battery initially at $120^{\circ}C$ is placed in an environment at $30^{\circ}C$. After 10 minutes, its temperature drops to $80^{\circ}C$. Using Newton's law of cooling, **formulate**, **solve**, and **estimate** the temperature after 20 minutes. Also **interpret** system behavior.
23.	A heated metal rod cools from $200^{\circ}C$ to $150^{\circ}C$ in 5 minutes in a room at $25^{\circ}C$. **Find** the time required to reach $50^{\circ}C$ and **analyze** cooling efficiency.
24.	A body of mass 5 lb is dropped from a height of 100 ft with zero velocity. Assuming no air resistance, find a) an expression for the velocity of the body at any time t, b) an expression for the position of the body at any time t,
25.	A sensor cools from $90^{\circ}C$ to $70^{\circ}C$ in 8 minutes in a $20^{\circ}C$ room. **Find** the cooling constant k.
26.	A body is dropped from rest. **Formulate** the differential equation and **find** velocity.
27.	Apply Newton's law of cooling to formulate the differential equation for an object at $90^{\circ}C$ placed in a surrounding of $25^{\circ}C$.
28.	Apply the concept of free fall to construct the differential equation and determine the velocity of a body dropped from rest.
29.	A radioactive sample initially has 80 mg. After 3 hours, it reduces to 60 mg. Formulate, solve, and evaluate the mass remaining after 6 hours.
30.	A bacterial culture grows from 500 to 1500 in 4 hours. Develop the model and predict population after 6 hours.
31.	A vehicle slows down in water where deceleration is proportional to velocity. Initial velocity is 20 m/s. After 5 seconds, velocity becomes 10 m/s. Formulate, solve, and interpret velocity after 10 seconds.

32.	An investment of ₹10,000 grows to ₹15,000 in 5 years. Determine the growth constant and interpret doubling time.
33.	A drug concentration reduces to 75% in 2 hours. Analyze decay constant and estimate concentration after 5 hours.
34.	A motorboat slows down such that its retardation is proportional to velocity. **Formulate** the governing differential equation and **identify** its type.
35.	In a cooling system, **interpret** why temperature difference governs the rate of cooling.
36.	A viral infection spreads in a closed campus of 5000 students. Initially, 5 students are infected and after 4 days, 50 students are infected. Formulate the logistic model, determine constants, and evaluate infected population after 10 days.
37.	A fish population in a lake follows logistic growth with carrying capacity 2000. Initially, population is 100 and after 3 months it becomes 400. Develop the model and predict population after 8 months.
38.	A chemical reactor produces a substance limited by capacity 1000 units. Initially 50 units exist and after 2 hours it becomes 200 units. Formulate the logistic equation and analyze production after 6 hours.
39.	AIn a logistic system with carrying capacity 800, determine and interpret the population when growth rate is maximum.
40.	An engineer models growth as $dP/dt = kP^2$. Evaluate and correct the model using logistic principles.
41.	Explain why logistic growth is slower than exponential growth at later stages.
42.	Analyze why the logistic growth model initially behaves like exponential growth.
	Unit 2
1.	A spring-mass system is governed by $d^2y/dt^2 + 5dy/dt + 6y = 0$ Classify the equation, determine the complementary function, and interpret system behavior.
2.	An RLC circuit is modeled as $d^2i/dt^2 + 4di/dt + 4i = e^{(-t)}$ Classify the equation, determine the complementary function, and interpret system behavior.
3.	A control system follows $d^2y/dx^2 + 4dy/dx +13y = 0$ Analyze roots, determine CF, and interpret system stability.
4.	Classify and justify the equation: $d^2y/dx^2 + 3y = \sin x$
5.	A system follows $d^2y/dx^2 + 4y = 0$, with $y(0) = 2$, $y'(0) = -4$. The complementary function is $y = C_1\cos 2x + C_2\sin 2x$. Determine the constants and interpret system behavior.
6.	Interpret the role of particular integral in second order systems in physical and engineering contexts.

7.	Explain the role of initial conditions in second order systems.
8.	A spring-mass system is governed by $d^2y/dt^2 + 5dy/dt + 6y = 0$ Formulate the auxiliary equation, determine the complementary function, and analyze system behavior.
9.	An RLC circuit satisfies $d^2i/dt^2 + 4di/dt + 4i = 0$ Determine complementary function and evaluate system response.
10.	A control system is modeled by $d^2y/dx^2 + 2dy/dx + 10y = 0$ Analyze roots, determine complementary function, and interpret oscillation.
11.	Analyze the equation $d^2y/dt^2 + 6dy/dt + 9y = 0$ and interpret the nature of motion.
12.	A vibration system is modeled by $d^2y/dt^2 + 9y = 0$ Identify the nature of roots and interpret system behavior.
13.	Interpret why real negative roots indicate stable systems.
14.	Identify the condition for overdamped system in second order models.
15.	An RLC circuit is modeled by $d^2i/dt^2 + 3di/dt + 2i = e^{-t}$ Formulate, determine CF and PI using undetermined coefficients, and construct the complete solution.
16.	A vibration system is governed by $d^2y/dt^2 + 4y = t^2$ Determine particular integral using undetermined coefficients and evaluate complete response.
17.	A control system is described by $d^2y/dx^2 + y = \cos 2x$ Determine the particular integral using another method.
18.	An electrical system is described by $d^2i/dt^2 + 3di/dt + 2i = e^{(2t)}$ Find the particular integral.
19.	A vibrating system is governed by $d^2y/dt^2 + 4y = \sec t$ Evaluate and compare the applicability of undetermined coefficients and variation of parameters.
20.	A vibration system is modeled by $d^2x/dt^2 + 3dx/dt + 2x = 5\cos t$ Identify and justify whether the system is homogeneous or non-homogeneous.
21.	In a forced mechanical system, interpret the significance of complete solution.

22.	A mass of 2 kg is attached to a spring with stiffness 50 N/m. Formulate the differential equation, determine angular frequency, and evaluate the motion equation if initial displacement is 0.3 m and initial velocity is 0.
23.	A spring system follows $x'' + 16x = 0$ Analyze amplitude, frequency, and evaluate total energy expression.
24.	An LC circuit is governed by $d^2q/dt^2 + 9q = 0$ Determine the solution and interpret periodic behavior.
25.	A mass-spring system satisfies $x'' + 25x = 0$ Analyze and interpret amplitude, frequency, and nature of motion.
26.	An LC circuit has equation $d^2q/dt^2 + (k/m)q = 0$ Evaluate the effect of increasing k on frequency.
27.	A spring-mass system in vibration analysis is modeled by a second-order differential equation is $\frac{d^2x}{dt^2} + \omega^2 x = 0$. Interpret the type of motion.
28.	A mechanical oscillator completes 50 cycles in 10 seconds. Determine its frequency and interpret its engineering significance.
29.	A vehicle suspension system is modeled by $m\, d^2x/dt^2 + c\, dx/dt + kx = 0$ with m = 2 kg, c = 6 Ns/m, k = 5 N/m. Formulate the differential equation, determine the nature of damping, and interpret system response.
30.	A vibration system satisfies $d^2x/dt^2 + 8dx/dt + 16x = 0$ Analyze the damping condition and evaluate system response.
31.	A system is governed by $d^2x/dt^2 + 10dx/dt + 21x = 0$ Determine the nature of roots and interpret system behavior.
32.	A system has equation $d^2x/dt^2 + 4dx/dt + 13x = 0$ Classify the system and interpret its motion.
33.	Interpret the role of damping co-efficient in damped vibration systems used in engineering structures.
34.	A system has equation $d^2x/dt^2 + 4dx/dt + 8x = 0$ Classify the system and interpret its motion.

35.	Interpret overdamped system behavior in engineering systems..
36.	A spring-mass-damper system is subjected to an external force $F(t) = 10\cos 2t$. Given m = 1 kg, c = 4 Ns/m, k = 5 N/m, formulate the differential equation, determine steady-state solution, and interpret system behavior.
37.	A structure vibrates under external force $F(t) = 8\cos\omega t$. Analyze and determine the condition for resonance and evaluate its effect.
38.	A forced system follows $x'' + 2x' + 10x = 5\cos 3t$ Determine amplitude response and analyze phase difference.
39.	Analyze and interpret the difference between transient and steady-state response in forced vibration systems.
40.	A spring-mass-damper system is governed by $x'' + 4x' + 5x = 10\cos 2t$ Determine the steady-state amplitude and interpret system behavior.
41.	A forced system satisfies $x'' + 6x' + 25x = 10\cos 4t$ Determine phase difference.
42.	A system has parameters k = 16, c = 2, $\omega = 3$, $F_0 = 6$. Compute steady-state amplitude.
	Unit 3
1.	An electric vehicle follows the trajectory $\vec{r}\left(t\right) = t^2\hat{i} + 3t\hat{j}$. Compute velocity and acceleration at $t = 2$. Analyze whether the motion is uniformly accelerated and justify.
2.	A robotic system path is given by $\vec{r}\left(t\right) = \cos t\,\hat{i} + \sin t\,\hat{j}$. Compute velocity and unit tangent vector at $t = \pi/2$. Interpret direction of motion.
3.	A drone path is $\vec{r}\left(t\right) = t^3\hat{i} + 2t^2\hat{j} + t\hat{k}$. Compute velocity, acceleration, and evaluate tangential component at $t = 1$.
4.	In a conveyor system, $\vec{r}\left(t\right) = t^2\hat{i} + t^2\hat{j}$. Analyze whether direction of velocity changes with time.
5.	A student computes derivative of $\vec{r}\left(t\right) = t^2\hat{i} + e^t\hat{j}$ incorrectly. Identify error if $\vec{v} = 2t\hat{i} + te^t\hat{j}$ and correct it.
6.	A particle has $\vec{r}\left(t\right) = 4t\hat{i} + 3t\hat{j}$. Compute speed and interpret motion.
7.	A system models only acceleration without velocity. Identify limitation.
8.	In an electric vehicle battery system, temperature distribution is modeled by $T\left(x, y, z\right) = x^2 + y^2 + z^2$. Compute the gradient and evaluate the direction of maximum temperature rise at point (1,1,1). Interpret its physical significance for thermal management and justify.

9.	In a heat sink design, temperature field is $T\left(x, y\right) = 4x^2 + 2y^2$. Compute directional derivative at (1,1) in direction $\vec{v} = 3\hat{i} + 4\hat{j}$. Analyze effectiveness of heat dissipation in that direction.
10.	In an electrostatic field, potential $V\left(x, y, z\right) = xyz$. Compute gradient and evaluate directional derivative at (1,1,1) along $\hat{i} + \hat{j} + \hat{k}$. Interpret physical meaning.
11.	In a thermal system, $T\left(x, y\right) = x^2 + y^2$. Identify direction where no temperature change occurs and justify using gradient concept.
12.	An engineer assumes gradient gives minimum increase direction. Identify the error and correct interpretation.
13.	For $T\left(x, y\right) = 3x + 4y$, compute gradient and interpret field behavior.
14.	A system uses non-unit vector for directional derivative. Identify issue.
15.	An electric vehicle moves along a path $\vec{r}\left(t\right) = 2t\,\hat{i} + 3t^2\,\hat{j}$. Compute velocity and acceleration at $t = 2$ and analyze the nature of motion.
16.	A robotic arm position is given by $\vec{r}\left(t\right) = t^2\,\hat{i} + 3t\,\hat{j} + 4\hat{k}$. Evaluate velocity and justify whether speed is constant.
17.	A robotic arm follows the path $\vec{r}\left(t\right) = t\hat{i} + t^2\hat{j}$. Compute the unit tangent vector at $t = 1$ and analyze the direction of motion. Justify its significance in motion control.
18.	A rocket follows the path $\vec{r}\left(t\right) = 4t\,\hat{i} + 2t^2\,\hat{j} - 3t^3\,\hat{k}$. Find velocity and acceleration at $t = 2$ and interpret the motion.
19.	A drone trajectory is $\vec{r}\left(t\right) = \cos t\,\hat{i} + \sin t\,\hat{j}$. Compute tangent and normal vectors at $t = 0$ and interpret curvature.
20.	A fluid velocity field is given by $\vec{v}\left(t\right) = 3t\,\hat{i} + 2t^2\,\hat{j}$. Identify whether acceleration is constant.
21.	A vehicle moves along $\vec{r}\left(t\right) = t^2\hat{i} + t^3\hat{j}$. Compute curvature at $t = 1$ and evaluate how direction changes affect steering control.
22.	In a conveyor system, path $\vec{r}\left(t\right) = t\hat{i} + t^2\hat{j}$. Identify if motion is straight or curved using tangent direction.
23.	An engineer models position as a scalar $x\left(t\right) = t^2$instead of a vector. Identify the error and correct it.
24.	An engineer assumes tangent and normal vectors are parallel. Identify and correct the error.

25.	Given $\vec{r}\left(t\right) = 3t\,\hat{i} + 2t^2\,\hat{j}$, identify the type and justify.
26.	Temperature is given by $T\left(x, y\right) = 5x + 3y$. Identify the type of field.
27.	A particle moves in circle. Identify curvature nature.
28.	Identify role of unit tangent vector in motion analysis.
29.	In a fluid cooling system of an electric vehicle, the velocity field is given by $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$. Compute the divergence and analyze whether the flow represents a source or incompressible system. Justify your conclusion.
30.	In an electromagnetic field, the vector field is $\vec{F} = 2x\hat{i} + 3y\hat{j}$. Compute divergence and evaluate whether the field conserves flux. Interpret physically.
31.	A pipe flow system is modeled by $\vec{F} = yz\hat{i} + xz\hat{j} + xy\hat{k}$. Compute divergence and analyze whether the system satisfies incompressibility.
32.	In a heat transfer system, $\vec{F} = x^2\hat{i} + y^2\hat{j}$. Identify divergence and interpret whether heat accumulates.
33.	An engineer assumes divergence is always zero for any vector field. Identify and correct the error.
34.	Compute divergence of the electric model $\vec{F} = 3x\hat{i} + 4y\hat{j}$.
35.	Identify condition for incompressible fluid flow.
36.	In a fluid mixing system, velocity field is $\vec{F} = y\hat{i} - x\hat{j}$. Compute curl and analyze whether the flow is rotational. Interpret its impact on mixing efficiency.
37.	In an electromagnetic system, field $\vec{F} = 2x\hat{i} + 3y\hat{j} + 4z\hat{k}$. Compute curl and evaluate whether the field is conservative. Justify.
38.	A projectile in a mechanical system is launched with velocity $\vec{u} = 10\hat{i} + 20\hat{j}$. Compute time of flight and maximum height. Analyze motion using vector formulation.
39.	In a fluid flow, $\vec{F} = x\hat{i} + y\hat{j}$. Identify curl and interpret rotational behavior.
40.	An engineer claims all zero-curl fields are non-conservative. Identify and correct the error.
41.	Compute curl of constant field $\vec{F} = 5\hat{i} + 3\hat{j}$
42.	Interpret the physical meaning of curl in fluid flow.
	Unit 4

1.	A particle moves under force field F = (2x + y)i + (x − y)j along the straight line from (0,0) to (1,1). Formulate and evaluate the work done using line integral.
2.	A field is given by F = (2x)i + (2y)j Analyze whether the field is conservative and evaluate work done from (0,0) to (1,1).
3.	A force field is F = (y)i + (x)j Evaluate work done along two paths from (0,0) to (1,1): (i) y = x (ii) $y = x^2$ Analyze path dependence.
4.	A force field acts on a particle moving along a curve. Analyze and interpret the physical meaning of the line integral.
5.	A robot follows path $r(t) = ti + t^2j$ Determine velocity vector and interpret motion.
6.	Differentiate scalar and vector line integrals.
7.	Define path independence in engineering systems.
8.	Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ for the **fluid velocity field** $\vec{F} = 2xy\,\hat{i} + yz\,\hat{j} + xz\,\hat{k}$, where $\hat{n} = \hat{i}$ and the surface is $x = 2,\ 0 \le y \le 1,\ 0 \le z \le 2.$
9.	Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ for the **heat flux field** $\vec{F} = x^2\,\hat{i} + yz\,\hat{j} + z^2\,\hat{k}$, where $\hat{n} = \hat{j}$ and the surface is $y = 1,\ 0 \le x \le 2,\ 0 \le z \le 1.$
10.	Evaluate $\iint_S \vec{F}\cdot\hat{n}\,ds$ for the **mass flow field** $\vec{F} = xyz\,\hat{i} + y^2\,\hat{j} + xz\,\hat{k}$, where $\hat{n} = \hat{k}$ and the surface is $z = 1,\ 0 \le x \le 1,\ 0 \le y \le 2.$
11.	Determine the type of flow for $\vec{F} = x\hat{i} + y\hat{j}$ at the point $(1, 2)$ across a surface with $\hat{n} = \hat{i}$.
12.	Evaluate the direction of flow for $\vec{F} = y\hat{j} + z\hat{k}$ at point $(1, 2, 3)$ across a surface with $\hat{n} = \hat{j}$.
13.	Determine the nature of flow for the vector field $\vec{F} = 3\hat{i} + 2\hat{j}$ across a surface with unit normal $\hat{n} = \hat{i}$.
14.	Determine the nature of flow for $\vec{F} = -4\hat{j} + \hat{k}$ across a surface with $\hat{n} = \hat{j}$.
15.	Evaluate and interpret the total outward heat flux using divergence theorem for $F = 4xz\,i - y^2\,j + yz\,k$ over a unit cubic control volume (0 ≤ x,y,z ≤ 1)
16.	Evaluate and interpret the total outward heat flux using divergence theorem for $F = (x^2 - yz)i + (y^2 - xz)j + (z^2 - xy)k$ over a rectangular domain 0 ≤ x ≤ a, 0 ≤ y ≤ b, 0 ≤ z ≤ c.
17.	Evaluate the outward flux of $F = 4x\,i - 2y^2\,j + z^2\,k$ through a cylindrical pipe ($x^2 + y^2 = 4$, 0 ≤ z ≤ 3). Interpret system behavior.
18.	Determine whether the vector field $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ has sources or sinks using divergence.

19.	Use Gauss theorem to compute total outward **heat flux** for $\vec{F} = 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$ over a unit cube.
20.	State Gauss divergence theorem and mention its significance in fluid flow analysis.
21.	For a **fluid velocity field** $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$, compute divergence and interpret its meaning.
22.	In a **fluid flow system**, the velocity field is given by $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$. Find the **circulation (work done)** of the fluid around the boundary of the region in the **xy-plane (z = 0)** bounded by $0 \le x \le 1,\ 0 \le y \le 1$ using Stokes' theorem.
23.	In an **electromagnetic field**, the vector field is $\vec{F} = z\hat{i} + x\hat{j} + y\hat{k}$. Determine the **circulation** around the boundary of the region in the **yz-plane (x = 0)** bounded by $0 \le y \le 1,\ 0 \le z \le 1$ using Stokes' theorem.
24.	In a **heat transfer system**, the flux field is $\vec{F} = xy\hat{i} + yz\hat{j} + zx\hat{k}$. Compute the **circulation of heat flow** around the boundary of the region in the **xz-plane (y = 0)** bounded by $0 \le x \le 1,\ 0 \le z \le 1$ using Stokes' theorem.
25.	In a **fluid system**, $\vec{F} = y\hat{i} - x\hat{j}$. For the region in the **xy-plane** bounded by $0 \le x \le 1,\ 0 \le y \le 1$0≤x≤1,0≤y≤1, compute $\left(\nabla \times \vec{F}\right) \cdot \hat{n}$ using Stokes' theorem.
26.	In an **electromagnetic field**, $\vec{F} = z\hat{j} - y\hat{k}$. For the region in the **yz-plane** bounded by $0 \le y \le 1,\ 0 \le z \le 1$, compute $\left(\nabla \times \vec{F}\right) \cdot \hat{n}$ using Stokes' theorem.
27.	In a **fluid flow model**, state Stokes' theorem and identify the direction of $\hat{n}$ for a surface in the **xy-plane.**
28.	In a **fluid flow model**, state Stokes' theorem and identify the direction of $\hat{n}$ for a surface in the **xz-plane**.
29.	In a **fluid circulation system**, the velocity field is $\vec{F} = \left(x^2 - y\right)\hat{i} + \left(x + y^2\right)\hat{j}$. Use Green's theorem to **evaluate** the circulation around the rectangular region bounded by $0 \le x \le 1,\ 0 \le y \le 2$.
30.	In a **heat transfer system**, the heat flow is given by $\vec{F} = \left(xy\right)\hat{i} + \left(x^2 + y\right)\hat{j}$. Use Green's theorem to **analyze** the net circulation (work done) around the region bounded by $x = 0,\ y = 0,\ x + y = 1$.
31.	In an **electrical field model**, $\vec{F} = \left(y^2\right)\hat{i} + \left(x^2\right)\hat{j}$. Use Green's theorem to **evaluate** the work done around the square bounded by $0 \le x \le 1,\ 0 \le y \le 1$.
32.	In a **fluid flow model**, $\vec{F} = y\hat{i} + x\hat{j}$. Use Green's theorem to **compute** $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ and interpret its meaning.
33.	In a **heat flow system**, $\vec{F} = \left(x + y\right)\hat{i} + \left(x - y\right)\hat{j}$. **Analyze** the integrand in Green's theorem.

34.	In a **fluid flow system**, $\vec{F} = \left(x + y\right)\hat{i} + \left(2x - y\right)\hat{j}$. Compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$∂x∂Q−∂y∂P.
35.	In a **circulation model**, what does a positive value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ indicate?
36.	In a **fluid transport system**, the velocity field is $\vec{F} = \left(2x\right)\hat{i} + \left(y\right)\hat{j} + \left(z\right)\hat{k}$. Use a **surface integral** to compute the total **flux** through the surface $x = 1$ bounded by $0 \le y \le 2,\ 0 \le z \le 1$.
37.	In an **electrical field**, $\vec{F} = \left(y\right)\hat{i} + \left(x\right)\hat{j}$. Use a **line integral** to determine the **work done** along the boundary of the square defined by $0 \le x \le 1,\ 0 \le y \le 1$.
38.	In a **heat flow system**, $\vec{F} = \left(xz\right)\hat{i} + \left(yz\right)\hat{j} + \left(xy\right)\hat{k}$. Evaluate the **flux** through the surface $z = 1$z=1 bounded by $0 \le x \le 1,\ 0 \le y \le 2$.
39.	In a **fluid system**, $\vec{F} = y\hat{i} + z\hat{j} + x\hat{k}$. Compute $\vec{F} \cdot \hat{n}$ for the surface $y = 1$.
40.	In a **heat transfer model**, $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$. Determine whether flux $(\vec{F} \cdot \hat{n})$ through $z = 0$ surface exists.
41.	In a **fluid flow system**, the velocity field is $\vec{F} = x\hat{i} + y\hat{j} + z\hat{k}$. Find $\vec{F} \cdot \hat{n}$ on the surface $x = 1$.
42.	In a **heat transfer model**, what does a positive surface integral indicate?
	Unit 5
1.	In a two-dimensional incompressible fluid flow system, the velocity potential and stream function are modeled by $u(x,y) = x^2 - y^2$, $v(x,y) = 2xy$. Analyze whether the complex function $f(z) = u + iv$ satisfies the Cauchy–Riemann equations.
2.	An electromagnetic potential field is represented by $f(z) = (x^2 - y^2 + x) + i(2xy - y)$. Evaluate whether the function satisfies the Cauchy–Riemann equations.
3.	A signal propagation model is given by f(z) = e^x(cos y + i sin y). Determine whether the function is analytic using the Cauchy–Riemann equations and interpret the result.
4.	A control response is modeled by $f(z) = z^2$. Analyze whether the function is differentiable at $z = 1 + i$.
5.	A communication signal is represented by a complex function f(z). Evaluate the relationship between differentiability and continuity in complex systems.
6.	A signal transformation is defined by $w = z^2$. Identify whether the function is single-valued or multi-valued and justify.

7.	An impedance transformation is represented by $w = \sqrt{z}$. Analyze whether the function is single-valued or multi-valued.
8.	In a two-dimensional incompressible fluid flow, the velocity potential is given by $\phi(x,y)=x^2-y^2+x$. Construct the complex potential using the Milne–Thomson method and interpret the flow behavior.
9.	An electric potential field is represented by $u(x,y)=x^2-y^2-2y$. Determine the harmonic conjugate and construct the analytic function.
10.	A velocity potential in an aerodynamic flow field is u(x,y)=e^x(x cosy−y siny). Verify harmonicity and construct the corresponding complex potential.
11.	In a fluid transport system, the stream function is given by $v(x,y)=2xy$. Determine the harmonic conjugate using the Milne–Thomson method.
12.	An electric field potential is represented by u(x,y)=e^x cosy. Construct the analytic function using the Milne–Thomson method.
13.	In a thermal distribution model, the temperature potential is represented by $f(z) = (x^2 - y^2 + 2x) + i(2xy + 2y)$. Verify whether the function is analytic using the Cauchy–Riemann equations and interpret the physical meaning.
14.	A wave propagation model is represented by $u(x,y) = x^3 - 3xy^2$. Determine whether u is harmonic and analyze its engineering significance.
15.	A communication signal is modeled by f(z) = e^z. Analyze whether the function is analytic and interpret its significance in signal amplification.
16.	Evaluate the singular point of the function $f(z) = 1/(z - 2i)$ and interpret its engineering significance.
17.	State the condition for a harmonic function in engineering analysis.
18.	In a **fluid flow model**, the imaginary part of an analytic function is $v(x,y) = x^2 - y^2$. Using Milne–Thomson method, find $f'(z)$. Hint: $f'(z) = v_y(z,0) + iv_x(z,0)$
19.	In an **electrical field application**, the real part of an analytic function is $u(x,y) = x^2 + 2xy$. Find $f'(z)$ using Milne–Thomson method. Hint: $f'(z) = u_x(z,0) - iu_y(z,0)$

20.	In a two-dimensional incompressible fluid flow, the velocity potential is given by $u(x,y) = x^2 - y^2 + x$. Construct the corresponding analytic function using the Milne–Thomson method and interpret the engineering significance.
21.	In an **electrical field system**, the imaginary part is $v(x,y) = 2xy$. Construct the analytic function $f(z)$.
22.	In a **heat transfer application**, determine whether $u\left(x,y\right) = x^2 - y^2$is harmonic and construct its harmonic conjugate.
23.	In a **fluid flow system**, verify whether $u\left(x,y\right) = x^3 - 3xy^2$is harmonic.
24.	In an **electrical potential model**, the real part is $u(x,y) = e^x \cos y$Find the harmonic conjugate $v(x,y)$.
25.	For $u\left(x,y\right) = x^2 + y^2$determine whether u is harmonic.
26.	A rectangular fluid flow region bounded by x = 0, y = 0, x = 2 and y = 1 is transformed under the mapping $w = z + (3 - 2i)$. Determine the transformed region in the w-plane and analyze the geometrical effect of the mapping.
27.	An electrostatic field boundary represented by $\|z\| = 2$ is transformed under $w = z + 1 + 4i$. Determine the image of the boundary and interpret the transformation.
28.	A signal transformation system uses the mapping $w = (1+i)z$. Determine the image of the region $y > 2$ in the w-plane and analyze the transformation characteristics
29.	A circular airflow boundary $\|z\| = 3$ is transformed under $w = 4z$. Determine the image boundary and analyze the scaling effect.
30.	Analyze whether the mapping $w = z^2 + 2$ is conformal at $z = 1+i$.
31.	Determine the critical point of the transformation $w = z^2 - 6z + 5$.
32.	In a **fluid flow system**, the complex function is $f\left(z\right) = z^2$. Determine the critical point of the mapping.
33.	A signal transformation system maps the points $z_1 = 0$, $z_2 = i$, $z_3 = -i$ onto $w_1 = 1$, $w_2 = -1$, $w_3 = 0$. Determine the bilinear transformation and interpret the engineering significance.

34.	A control transformation maps the points $z_1 = 1, z_2 = i, z_3 = -1$ onto $w_1 = 0, w_2 = 1, w_3 = -1$. Determine the bilinear transformation.
35.	An electromagnetic field mapping transforms $z_1 = 2, z_2 = 0, z_3 = -2$ onto $w_1 = 1, w_2 = i, w_3 = -1$. Determine the corresponding bilinear transformation.
36.	Determine the fixed points of the bilinear transformation $w = \frac{3z-2}{z}$ and analyze their significance.
37.	Evaluate the fixed point of the transformation $w = \frac{2z+1}{z+2}$
38.	Identify the condition for invariant points in a bilinear transformation.
39.	Interpret the engineering significance of fixed points in bilinear mappings.
40.	In a **steady-state heat distribution system**, the temperature function is $u(x, y) = 3x - 2y + 5$. Evaluate whether the function is harmonic.
41.	In a **fluid potential model**, the functions are $u\left(x, y\right) = x^2 - y^2, v\left(x, y\right) = 2xy + y$Analyze whether an analytic function exists.
42.	In a **fluid flow and electrical potential model**, state the conditions required for the construction of an analytic function $f(z) = u + iv$.